

12/3 Complex Algebraic surface

□

Fiber bundle v.s. fibration for surface

decompose surf into lower-dim base & fiber.

reducing global problem to simple problem on curves.

Def Fiber bundle (= locally triv.) $f: X \rightarrow B$

(analytic/algebraic)

Zariski) open $\{U_i\}$ of B $f^{-1}(U_i) \xrightarrow{\varphi_i} U_i \times F$

analytic

$$\text{w/ } \varphi_i, \varphi_j^{-1}: (U_i \cap U_j) \times F \rightarrow (U_i \times U_j) \times F$$

$$t_{ij}: (U_i \cap U_j) \rightarrow G \subseteq \text{Aut}(F)$$

$$t_{ik} = t_{ij} t_{jk}$$

Fibration (= possibly with sing fiber / variation in family)

$f: X \rightarrow B$ X : conn sm surf, B : sm conn curve

proper, surj, conn fiber

"
sep, fin, univ closed. ($f^{-1}(b)$: cpt)

Q subadditivity F : gen fiber

$$|k(X) \geq k(F) + k(B)|$$

Complexity at least the combined complexity of
"horizontal" + "vertical"

Recall k $-\infty$ 0 1

model

$\mathbb{P}^1$

$g=1$

$g \geq 2$

moduli

pt

$\mathbb{C}$

$\mathcal{M}_g(3g-3)$

$$\square F = \mathbb{P}^1$$

o $\mathbb{P}$ -fibration = ruled surf $\sim_{\text{bir}} \mathbb{C} \times \mathbb{P}^1$
[Bea] III.4

$\hookrightarrow \mathbb{C} = \mathbb{P}^1$: rational surf

o $\mathbb{P}^1$ -bundle = geom. ruled surf $\simeq \mathbb{P}_{\mathbb{C}}(E)$ E : vect bundle of rank 2
[BHV] Prop 4.1

eg $\mathbb{P}^2$ -fib

[2]

o del Pezzo (2-dim Fano)

$$\mathbb{P}^2, \underbrace{\mathbb{P}^1 \times \mathbb{P}^1, Bl_1 \mathbb{P}^2, Bl_2 \mathbb{P}^2, \dots, Bl_r \mathbb{P}^2}_{\mathbb{P}^1\text{-bundle}}$$

minimal

$\mathbb{P}^1$ -bundle = classification of vect bundle of rk 2

$B = \mathbb{P}^1$ [Grothendieck] (decomposable)

$$E = \mathcal{O} \oplus \mathcal{O}(n) \text{ (up to } L)$$

$\Rightarrow F_n := \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ Hirzebruch surf

$$\underbrace{F_0 = \mathbb{P}^1 \times \mathbb{P}^1}_{\text{min}}, \underbrace{F_1 = Bl_1 \mathbb{P}^2}_{\text{not min}}, \underbrace{F_2 \dots}_{\text{min}}$$

$B = \text{ellip}$ [Atiyah]

$$\deg(E \otimes L) = \deg E + 2 \deg L$$

- split $L_1 \oplus L_2$

- Atiyah bundle F_r of deg 0

$$0 \rightarrow \mathcal{O} \rightarrow F_2 \rightarrow \mathcal{O} \rightarrow 0$$

- $F_{2,1}$ of deg 1 $0 \rightarrow \mathcal{O} \rightarrow F_{2,1} \rightarrow \mathcal{O}(p) \rightarrow 0$

$B = \text{gen}$ [Tyurin]

Fact III.10 C: irrational $\Rightarrow$ geom. ruled. $\mathbb{P}_C(V)$: min.

Review (Kodaira - Enriques classification)

k		P_g	g	Alg
$-\infty$	min rat surf	0	0	Y
	ruled / g	0	g	Y
	VII ₀	0	0	N
0	K3	1	0	Y/N
	complex tori / Ab.	1	2	Y/N
	Enriques	0	0	Y
	biellip	0	1	Y
	Primary Kodaira	1	1	N
	Secondary Kodaira	0	1	N

$$b_2 = 3, b_1 = 1$$

1	ellip surf	-	-	$\mathbb{Y}/N$	(3)
2	gen type	-	-	$\mathbb{Y}$	

Elliptic - bundle

$$\text{Aut}(E) = E \times \mathbb{Z}/n\mathbb{Z}$$

translation rotation fixing 0

$$\left\{ \begin{array}{l} n=4 \quad \text{if } E = \mathbb{C}/\mathbb{Z} \oplus \mathbb{Z}i \\ n=6 \quad \text{if } E = \mathbb{C}/\mathbb{Z} \oplus \mathbb{Z}\omega \\ n=2 \quad \text{otherwise} \end{array} \right.$$

$$X/U \cong U \times E \quad \text{twisted patching}$$

Aut(E)-valued cycle

ellip fib bundle = associated bundle of principal Aut(E)-bundle (torsor)

$$= H^1(B, \text{Aut}(E))$$

$$X = P \times_{\text{Aut}(E)} E$$

$$\textcircled{a} \quad H^1(B, E) \longrightarrow H^1(B, \text{Aut}(E)) \longrightarrow H^1(B, \mathbb{Z}/n\mathbb{Z})$$

principal E-bundle ↖ sheaves of germs of local hol maps
(str gp reduced to E)

$$\textcircled{a} \quad \text{write } \Lambda/E = \mathbb{C}/\Lambda \quad \xrightarrow{\quad} \quad E\text{-torsor}$$

$$H^1(B, \Lambda) \longrightarrow H^1(\mathbb{C}_B) \longrightarrow H^1(E) \xrightarrow{\quad \zeta \quad} H^2(B, \Lambda) \longrightarrow 0$$

$\zeta \longmapsto c(\zeta)$

chern class
(characteristic class)

$$1) \quad B = \mathbb{P}^1 \quad (\text{simply conn?})$$

$$\Rightarrow H^1(B, \mathbb{Z}/n\mathbb{Z}) = 0$$

then Aut(E) reduced to E

$$\begin{cases} c(\zeta) = 0 \Rightarrow \text{top. triv.} \\ c(\zeta) \neq 0 \Rightarrow \text{not top prod.} \end{cases}$$

$$\text{Thm 5.4} \quad \begin{cases} \Rightarrow \int c(\zeta) = 0 \Rightarrow \text{ellip} \times \mathbb{P}^1 \\ c(\zeta) \neq 0 \Rightarrow \text{Hupf surf } (V_{110}) \quad b_1 = 3. \end{cases}$$

$$2) \quad B = \text{ellip}$$

principal bundle 2-1) $c(\zeta) = 0 \Rightarrow \text{complex torus } (E_1 \times E_2 \text{ type})$

2-2) $c(\zeta) \neq 0 \Rightarrow \text{primary Kodaira.}$
 $\rightarrow \text{secondary}$

Non-principal $\Rightarrow$ bielliptic (7-cases).

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Elliptic fibration (rel min.)

Data

- section
- Functional inv.
- topological inv.
- (multi fiber)

$\begin{array}{c} + \\ + \\ + \end{array}$

j :

Section $s: B \rightarrow X$ s.t. $f \circ s = \text{id}_B$

$s(b) \in X_b$: holomorphic, $\rightarrow$ choose an origin.
continuous.

$= k (=k(B))$ -rational
point of gen fib.

Multi section $Z \subset X$: closed sub sch.

s.t. $f|_Z: Z \rightarrow B$ fin & surj

$d := \deg(f|_Z)$

$\rightarrow$ choice simultaneously d -pts $\in X_b$.

L/k : fin ext.

$= L$ -rational point
of gen. fib.

$L = k(B')$ $B \rightarrow B'$ fib.

Note $Jac(f) = \mathbb{P}^1 \times_{\mathbb{P}^1} \mathcal{O}_X / \mathbb{P}^1 \times_{\mathbb{P}^1} \mathbb{Z}_X$ (see V.9).

Note $Jac(F) = H^1(B, \mathcal{O}_F) / H^1(B, \mathbb{Z}) \mid \{ \mathcal{O}_{X_b} \in \text{Pic}^0(X_b) \}$

X : $Jac(f)$ -torsor $H^1(B, Jac(f))$ (Weil-Châtelet)

j inv Weierstrass model.

$$y^2 = 4x^3 - g_2x - g_3$$

$$\Delta = g_2^3 - 27g_3^2 \neq 0 \quad (\Leftrightarrow \text{sm})$$

$$j(F) = \frac{(1728)g_2^3}{\Delta(F)}$$

g_2 : weight 4 modular

g_3 : weight 6 modular

Δ : weight 12 cusp

weight 0 modular $j = 0, 1$

$w \geq 2$

$T: V/H \rightarrow \mathbb{C}$

→ fibration with section $\mathbb{C} \times \Delta / \mathbb{Z} + \mathbb{Z}Z(s)$

[5]

$$J(s) = j(Z(s)) = \frac{g_2(s)^3}{g_2(s)^3 - 27g_3(s)^2}$$

$j: B \rightarrow P'$ j : holomorphic $\Rightarrow$ const $\Rightarrow$ isotrivial

j : non-isotrivial $\Rightarrow \exists$ pole!

• Monodromy



$$X_t \xrightarrow{\sim} X_t$$

$$A/A_t$$

$$A_t = A \cdot A_{t_0}$$

$$A \in SL_2(\mathbb{Z})$$

$j = \infty$: multiplicative-type
 I_b, I_b^*
 $j < \infty$: additive type or sm

$$R: \pi_1(B^0) \rightarrow SL_2(\mathbb{Z}) \quad B^0 = B \setminus j^{-1}(0, 1, \infty)$$

($\Leftrightarrow R^* \mathbb{Z} \times$ locally system VHS)

• Semistable reduction (Deligne - Schmidt)

taking fin over $t \mapsto t^m$

T : unip ($1 + \text{nilp}$)

$$\Rightarrow N := \log T = (T - I) - \frac{1}{2}(T - I)^2 + \dots \text{eventually } 0.$$

• $N = 0$ Type I

• $N \neq 0, N^2 = 0$ Type II

• $N^2 \neq 0, N^3 = 0$ Type III

Type I $\rightarrow$ sm

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

• Type II $\rightarrow$



$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

$$Z(s) = Z(0) + \frac{b}{2\pi i} \log s$$

• Type III $\rightarrow$ impossible (2-dim)

possibilities of sing. fib [Kodaira]

(6)

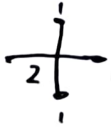
Idea: $\chi_0 = \sum n_i C_i$ use $\chi_0^2 = 0$, $K_X \cdot \chi_0 = 0$

$P_a(\chi_0) = 1$, adjunction formula

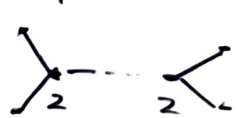
Zariski's lemma $(C_i, G) < 0$
euler number

$e(F) = \# \text{vertex-edge}$
for SNC

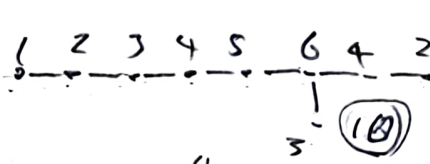
I_0

$I_0^* \tilde{P}_4$  (6)

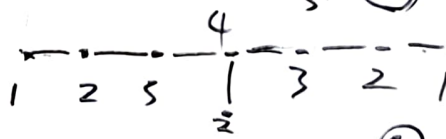
I_1  rat. curve w/ node

$I_1^* \tilde{D}_{4+b}$  (b+6)

I_b $\tilde{A}_{b-1}$  (b)

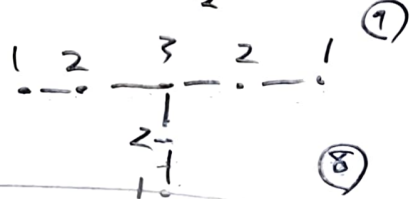
$II^*(\tilde{F}_8)$  (10)

II $<$ rat curve w/ cusp (2)

$III^*(\tilde{F}_4)$  (7)

III $\tilde{A}_1$  (3)

IV $\tilde{A}_2$  (4)

$IV^*(\tilde{F}_6)$  (8)

in $I_0, m I_b$. (non simply conn.)

$\subset \exists$ free fm. gp action.

$$e(X) = \sum e(F_i)$$

Type 1



may

$\mathbb{Z}/6\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$

$E \times \Delta \hookrightarrow G$

Mitanda-Perron

$E \rightarrow E/G \cong \mathbb{P}^1$ Riemann Hurwitz

$(v; e_1, e_2, e_3) = (3; 2, 3, 6), (3; 2, 4, 4), (3; 3, 3, 3)$

ramification index

$$2g(F) - 2 = n(2g(\mathbb{P}^1) - 2) + \sum (e_i - 1)$$

eg



w/ $A_{6,1}, A_{3,1}, A_1$

or A_5, A_2, A_1

smpt on E/G

but sing pt on $E \times \Delta / G$

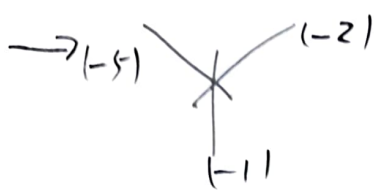
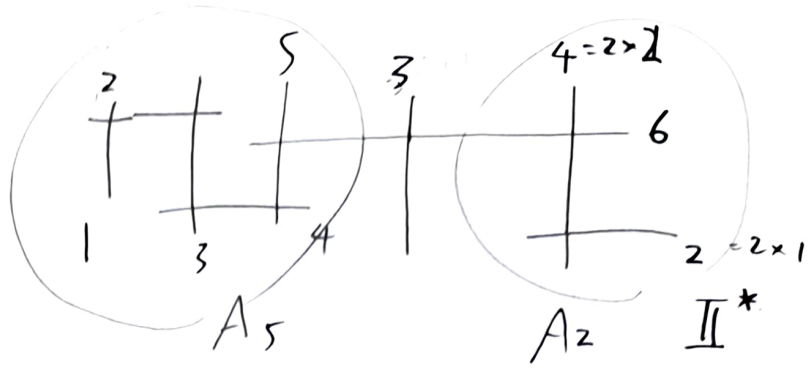
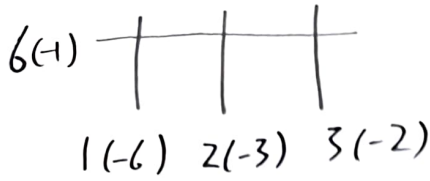
$$A_{r,a} \cong \mathbb{C}^{2/r} / \Gamma(1, a)$$

Hirzebruch-Jung

$$(x, y) \sim (\sum x_i, \sum y_i^a)$$

$$A_{r-1} = A_{r,r-1}$$

$$\frac{6}{1} = 6 = [6]$$

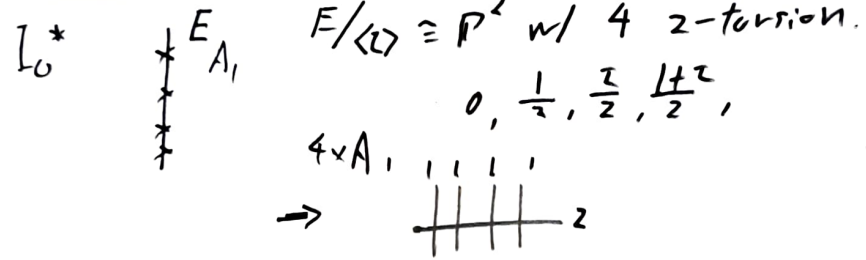


$$\frac{6}{5} = [2, 2, 2, 2, 2]$$



$$\frac{3}{2} = [2, 2]$$

$$\frac{2}{1} = [2]$$



Rem

$K3$ has ellip fib. $\Leftrightarrow \exists F \in NS, F^2 = 0, \text{ nef.}$

idea

$$f: X \rightarrow B$$

$X_b \sim f_*(b)$ as Cartier-div

$$b \sim b' \Rightarrow X_b \sim X_{b'} \quad F^2 = F \cdot F = X_b \cdot X_{b'} = 0.$$

Furques always admit ^{ellip} fiber

$$p \geq 5 \Rightarrow OK.$$

$$NS(X) \cong U \oplus F_5(-1)$$

all algebraic?

Canonical bundle formula _{moduli} $= \frac{1}{2} d^* \mathcal{O}_{\mathbb{P}^1(1)}$ _{boundary}

$$K_X = f^*(K_S \otimes (R^1 f_* \mathcal{O}_X)^\vee) \otimes \mathcal{O}_X(\sum (m_i - 1) F_i)$$

[BHPV]: Compact complex surfaces

III, V

[Bea]: Complex Algebraic surfaces

[Kon]: K3 surface

[Huy]: Lect on K3

$f: X \rightarrow \mathbb{C}$ elliptic surface w/ section

$L := (R^1 f_* \mathcal{O}_X)^\vee$ dual line bundle

$$\deg L = \chi(\mathcal{O}_X)$$

$$K_X = f^*(K_{\mathbb{C}} \otimes L)$$

$$X \text{ alg} \Leftrightarrow \chi(\mathcal{O}_X) = \deg L > 0$$

$$\text{and} \Rightarrow \underline{\text{isotrivial \& m } I_0 \text{ ?}}$$

$$\delta(f) := \chi(\mathcal{O}_X) + (2g(S) - 2) + \sum (1 - \frac{1}{m_i})$$

$$\delta(f) < 0 \Rightarrow k = -\infty$$

$$\delta(f) = 0 \Rightarrow k = 0$$

$$\delta(f) > 0 \Rightarrow k = 1$$